

Figures 2.2 and 2.3. In this case, the i th moment of X can be written as

$$\begin{aligned}
 m_i &= E(X^i) \\
 &= \int_{-\infty}^{\infty} x^i f_X(x) dx \\
 &= \int_{-\infty}^0 x^i f_X(x) dx + \int_0^{\infty} x^i f_X(x) dx
 \end{aligned} \tag{2.39}$$

If i is odd then $x^i = -(-x)^i$. Combined with the fact that $f_X(x) = f_X(-x)$, we see that

$$\begin{aligned}
 \int_{-\infty}^0 x^i f_X(x) dx &= \int_0^{\infty} (-x)^i f_X(-x) dx \\
 &= - \int_0^{\infty} x^i f_X(x) dx
 \end{aligned} \tag{2.40}$$

So for odd i , the i th moment in Equation (2.39) is zero. We see that all of the odd moments of a zero-mean random variable with a symmetric pdf are equal to 0.

2.3 TRANSFORMATIONS OF RANDOM VARIABLES

In this section, we will look at what happens to the pdf of an RV when we pass the RV through some function. Suppose that we have two RVs, X and Y , related to one another by the monotonic² functions $g(\cdot)$ and $h(\cdot)$:

$$\begin{aligned}
 Y &= g(X) \\
 X &= g^{-1}(Y) = h(Y)
 \end{aligned} \tag{2.41}$$

If we know the pdf of X [$f_X(x)$], then we can compute the pdf of Y [$f_Y(y)$] as follows:

$$\begin{aligned}
 P(X \in [x, x+dx]) &= P(Y \in [y, y+dy]) \quad (dx > 0) \\
 \int_x^{x+dx} f_X(z) dz &= \begin{cases} \int_y^{y+dy} f_Y(z) dz & \text{if } dy > 0 \\ - \int_y^{y+dy} f_Y(z) dz & \text{if } dy < 0 \end{cases} \\
 f_X(x) dx &= f_Y(y) |dy| \\
 f_Y(y) &= \left| \frac{dx}{dy} \right| f_X[h(y)] \\
 &= |h'(y)| f_X[h(y)]
 \end{aligned} \tag{2.42}$$

where we have used the assumption of small dx and dy in the above calculation.

²A monotonic function is a function whose slope is either always nonnegative or always nonpositive. If the slope is always nonnegative, then the function is monotonically nondecreasing. If the slope is always positive, then the function is monotonically increasing. If the slope is always nonpositive, then the function is monotonically nonincreasing. If the slope is always negative, then the function is monotonically decreasing.

■ EXAMPLE 2.4

In this example, we will find the pdf of a linear function of a Gaussian RV. Suppose that $X \sim N(\bar{x}, \sigma_x^2)$ and $Y = g(X) = aX + b$, where $a \neq 0$ and b are any real constants. Then

$$\begin{aligned}
 X &= h(Y) \\
 &= (Y - b)/a \\
 h'(y) &= 1/a \\
 f_Y(y) &= |h'(y)| f_X[h(y)] \\
 &= \left| \frac{1}{a} \right| \frac{1}{\sigma_X \sqrt{2\pi}} \exp \left\{ \frac{-[(y - b)/a - \bar{x}]^2}{2\sigma_X^2} \right\} \\
 &= \frac{1}{a\sigma_X \sqrt{2\pi}} \exp \left\{ \frac{-[y - (a\bar{x} + b)]^2}{2a^2\sigma_X^2} \right\}
 \end{aligned} \tag{2.43}$$

In other words, the RV Y is Gaussian with a mean and variance given by

$$\begin{aligned}
 \bar{y} &= a\bar{x} + b \\
 \sigma_Y^2 &= a^2\sigma_X^2
 \end{aligned} \tag{2.44}$$

This important example shows that a linear transformation of a Gaussian RV results in a new Gaussian RV.

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■ EXAMPLE 2.5

Suppose that we pass a Gaussian RV $X \sim N(0, \sigma_x^2)$ through the nonlinear function $Y = g(X) = X^3$:

$$\begin{aligned}
 X &= h(Y) \\
 &= Y^{1/3} \\
 h'(y) &= \frac{y^{-2/3}}{3} \\
 f_Y(y) &= |h'(y)| f_X[h(y)] \\
 &= \frac{y^{-2/3}}{3} \frac{1}{\sigma_x \sqrt{2\pi}} \exp[-x^2/(2\sigma_x^2)] \\
 &= \frac{y^{-2/3}}{3} \frac{1}{\sigma_x \sqrt{2\pi}} \exp[-y^{2/3}/(2\sigma_x^2)]
 \end{aligned} \tag{2.45}$$

We see that the nonlinear transformation $Y = X^3$ converts a Gaussian RV to a non-Gaussian RV. It can be seen that $f_Y(y)$ approaches ∞ as $y \rightarrow 0$. Nevertheless, the area under the $f_Y(y)$ curve is equal to 1 since it is a pdf.

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In the more general case of RVs related by the function $Y = g(X)$, where $g(\cdot)$ is a nonmonotonic function, the pdf of Y (evaluated at y) can be computed from the pdf of X as

$$f_Y(y) = \sum_i f_X(x_i) / |g'(x_i)| \tag{2.46}$$